



Effect of interstitial oxygen /nitrogen on mechanical and wear properties of TiZrHfNb refractory high-entropy alloy



Chi Jin, Xiaolin Li ^{*}, Junhong Kang, Haozhe Li, Haifeng Wang ^{*}

State Key Laboratory of Solidification Processing, Center of Advanced Lubrication and Seal Materials, Northwestern Polytechnical University, Xi'an, Shaanxi 710072, PR China

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ABSTRACT

Refractory high-entropy alloy (RHEA), as a kind of promising alloy, has received particular attention in high-temperature applications because of its stability and excellent high-temperature softening resistance. However, most RHEAs are in the strength-ductility trade-off dilemma, which leads to limited applications. In this study, the TiZrHfNb alloy with better ductility is selected as the base alloy, and the strength of the alloy is enhanced by interstitial solid-solution strengthening of O and N atoms, thereby improving the wear resistance of the alloy. All alloys exhibit a body-centered cubic (BCC) structure. The O-doped alloy exhibits a remarkable balance between strength and ductility, while the N-doped alloy displays the highest levels of hardness and yield strength, attributed to the grain boundary strengthening and solid-solution strengthening effects of interstitial atoms. The N-doped alloy has the lowest wear rate at room temperature and 500 °C, which is attributed to the hardening effect of N atoms. At 25 °C, the main wear mechanisms are severe abrasive wear, adhesive wear, and oxidative wear. At 500 °C, the dominant wear mechanisms change to oxidative wear and slight abrasive wear. The oxide layer on the worn surface can decrease the coefficient of friction (COF) and improve the wear resistance.

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1. Introduction

High-entropy alloys (HEAs) have attracted a great deal of attention as a novel class of alloys, which are usually composed of four or five elements in equiatomic or non-equiatomic ratios [1,2]. HEAs are considered a promising class of engineering materials that offer exciting possibilities for alloy design due to their exceptional mechanical properties [3–7], wear resistance [8–11], and corrosion resistance [12,13]. In recent years, a new strategy, doping interstitial elements in HEAs and acting as solid-solution strengthening mechanism, has been developed.

The mechanical properties of the equiatomic FeNiCoCrMn alloys were enhanced by the addition of 0.5 at% carbon, as reported by Wu et al. [14], which reduced the stacking fault energy and promoted the rapid formation of twins. Subsequently, Li et al. [15] reported that non-equiatomic Fe_{49.5}Mn_{29.7}Co_{9.9}Cr_{9.9}C₁ alloys demonstrate a remarkable combination of strength and ductility, due to the grain refinement strengthening and precipitation strengthening effects

induced by interstitial C atoms. In addition to C, other interstitial elements, such as N, O, and B, were also added in MEAs/ HEAs to optimize mechanical properties in recent years [6,16,17]. For example, Moravcik et al. [16] demonstrated that the addition of 0.5 at % N atoms to the CoCrNi MEA increased the yield strength without sacrificing too much ductility.

As a class of HEAs, refractory HEAs (RHEAs) are expected to be used in high-temperature fields due to high melting point and excellent softening resistance [18–20]. RHEAs are usually BCC structures with high yield strength. However, the lower ductility at room temperature limited their further development, compared with FCC HEAs [21,22]. Furthermore, lots of efforts have also been paid attention to the addition of interstitial elements in RHEAs, because the interaction between interstitial atoms and dislocations, and the alteration of deformation mechanisms can overcome the trade-off of strength and ductility [6,23–25]. It is particularly noted that Lei et al. [6] reported that oxygen atoms and other elements can form ordered oxygen complexes (OOCs) in the TiZrHfNb alloy, which can effectively hinder dislocation motion and change the plane slip to wave slip. The interaction between atoms and dislocations can achieve an excellent balance between dislocation pinning, multiplication, and substructure homogenization, thereby leading to an increase in both strength and ductility. In addition, Ye et. al [24] reported that the

^{*} Corresponding authors.

E-mail addresses: xiaolinli@nwpu.edu.cn (X. Li), haifengw81@nwpu.edu.cn (H. Wang).

addition of N and O elements in the TiZrHfNb alloy can increase the critical shear stress required to incipient plasticity at nanoscale, and emerge a large hardening effect. Interstitial elements can change the mechanical properties of HEAs remarkably. Moreover, the mechanical properties and tribological properties of alloys are inseparable [26]. Thus, it is valuable to study the effect of interstitial elements on the tribological properties of HEAs, especially the tribological properties of RHEAs at high temperatures.

The high-temperature friction and wear properties of RHEAs have recently been frequently reported [27–30]. Among them, TiZrHfNb RHEA has become an important candidate as a result of its excellent combination of strength and ductility. Furthermore, the addition of Al element in the TiZrHfNb RHEA can reduce the wear rate effectively at both room temperature and elevated temperature [28,29]. As far as we know, the inclusion of interstitial elements has the potential to enhance the mechanical properties of TiZrHfNb. Nevertheless, there exists a paucity of research that examines the tribological performance and elucidates the wear mechanism of TiZrHfNb with interstitial elements, particularly under high-temperature conditions.

In this study, the microstructure, mechanical properties, wear resistance, and mechanism of the TiZrHfNb, (TiZrHfNb)₉₈O₂, and (TiZrHfNb)₉₈N₂ alloys were systematically investigated and discussed from room temperature (25 °C) to 500 °C, to clearly elucidate the effect of O and N atoms, and also the relationship between the wear mechanism and service temperature.

2. Experimental

The three types of alloys with a nominal composition of TiZrHfNb, (TiZrHfNb)₉₈O₂, and (TiZrHfNb)₉₈N₂ (at%) were studied. Ti (99.99 wt%), Zr (99.95 wt%), Hf (99.95 wt%), and Nb (99.95 wt%) were chosen as the high-purity metals for the production of TiZrHfNb-based RHEA. The elements O and N were mixed into the TiZrHfNb-based alloy by the compounds TiO₂ (99.9 wt%) and TiN (99.9 wt%), respectively, hereafter denoted as O-2 and N-2 RHEA. The RHEAs' ingots were fabricated through vacuum arc-melting within a water-cooled copper mold, with each button-shaped alloy being flipped at least seven times to attain chemical homogeneity under a high-purity argon atmosphere. Prior to alloy melting, high-purity molten titanium acted as a getter to absorb residual oxygen. Subsequently, the molten alloys were subsequently cast into plates, with the as-cast ingots annealed for 24 h at 1200 °C and cold-rolled to reduce thickness by 20%. The rolled plate was then annealed for 30 min at 1200 °C to eliminate deformed microstructure and promote the formation of an equiaxed microstructure.

The mass fraction of the O and N elements, which was solved in TiZrHfNb RHEA, was evaluated using an oxygen and nitrogen analyzer (ON736, LECO, USA). X-ray diffractometry (XRD) was used to characterize the phase constitution of three alloys, utilizing a Bruker D8 Advance X-ray diffractometer. Cu-K α radiation ($\lambda = 1.54\text{\AA}$) was utilized in conjunction with a 2θ scattering range spanning 20–120°. The morphologies and chemical compositions of the worn surface were examined by FEI Helios G4 CX scanning electron microscopy (SEM) equipped with an Oxford electron backscatter diffraction (EBSD) system and energy-dispersive X-ray spectroscopy (EDS). The samples for the characterization of SEM and EBSD were electro-polished using a solution comprising 6% perchloric acid + 30% n-butyl alcohol + 64% methyl alcohol. The electro-polishing was

carried out for 30–40 s at a voltage of 35 V and a temperature of approximately 15 °C, using Struers LectroPol-5 equipment.

The CMT5105 universal tensile test machine was used to evaluate the tensile properties of the samples at room temperature. The strain rate was set at $2.0 \times 10^{-4} \text{ s}^{-1}$. Dog-bone-shaped tensile samples were prepared by electrical discharging, with a cross section of $1.0 \times 2.6 \text{ mm}^2$ and a gauge length of 15 mm. To ensure accuracy, at least 3 samples were tested for each alloy. In addition, hardness measurements were conducted using a Vickers hardness tester (MH-5 L) with a load of 5 N for 10 s. To obtain an average value, at least 15 indents were measured for each sample.

Reciprocating dry-sliding tests were performed using an RTEC MFT-5000 ball-on-disk wear machine at two different temperatures, 25 °C and 500 °C. The dimensions of the tribo-samples were $25 \times 6 \times 2 \text{ mm}^3$, obtained by cutting them from rolled plates, while the Si₃N₄ ball, possessing high thermal softening resistance and hardness, with a diameter of 9.5 mm was chosen as the counterpart. The sliding test was carried out with a normal load of 5 N and a sliding frequency of 5 Hz, to ensure clear visibility and easy measurement of the wear profile. The duration of the tests was 30 min, and the total sliding distance was 18 m. Before wear testing, the surfaces of the samples were ground to 2000 grit by SiC grinding papers. The chamber was preheated and left to stabilize for 10 min before conducting the reciprocating dry-sliding wear tests. The COF was recorded using the MFT software. The 3D profiles and cross-sectional curves, including the width and depth of the worn tracks, were obtained with a Bruker Contour GT-K 3D optical microscope after the wear tests. The wear volume was derived from the data obtained using Contour GT-K, and the specific wear rate (ω) was calculated using the following equation:

$$\omega = \frac{V}{L \times F} \quad (1)$$

where V is the wear volume (mm^3), L is the total sliding distance (m), and F is the applied normal load (N).

X-ray photoelectron spectroscopy (XPS) was employed to detect the chemical states of the worn surface using a PHI 5000 VersaProbe III instrument, with a monochromatic Al K α source at a beam size of 100 μm . The binding energy was corrected by setting the binding energy of the hydrocarbon C 1 s characteristic to 284.8 eV. The acquired peaks were fitted using PHI Multipak software and compared with those present in the NIST database.

Transmission electron microscopy (TEM) was employed to investigate the cross-sectional microstructures and chemical compositions of N-2 alloy after wear at 500 °C. Characterization was performed using bright-field and dark-field, high-resolution TEM, and high-angle annular dark-field scanning TEM (HAADF-STEM) techniques coupled with EDS, with an FEI Talos F200X TEM operated at 200 kV. The preparation of TEM samples was carried out using the FEI Helios G4 CX focused ion beam (FIB) milling technique.

3. Results and discussion

3.1. Microstructure and phase analysis

The XRD patterns of three RHEAs are shown in Fig. 1. A single-phase BCC structure is observed for all these RHEAs, as shown in Fig. 1a. The chemical composition of three RHEAs is listed in Table 1. The atomic percentages of O and N were accurately measured by the ON analyzer, and the atomic percentages of Ti, Zr, Hf, and Nb were

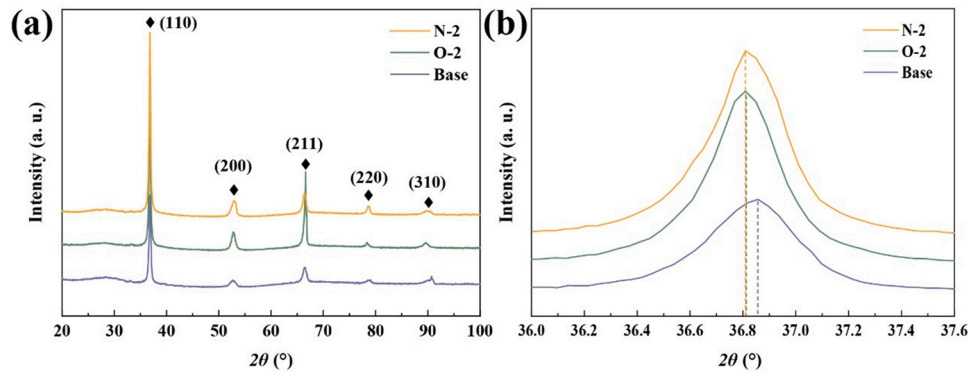


Fig. 1. (a) XRD patterns of the base, O-2, and N-2 RHEAs; (b) enlarged image of (110) diffraction peaks in (a).

obtained by EDS. By normalizing the atomic percentage of elements, the actual composition of three RHEAs is consistent with the design composition and no obvious segregation is observed. The enlarged view of the strongest diffraction peak (110) shows some certain offset due to the interstitial solid solution of O and N atoms. Therefore, the lattice parameters gradually increase with the addition of O and N atoms.

The microstructure morphologies, phase composition maps, and grain sizes of the base, O-2, and N-2 RHEAs are displayed in Fig. 2. Three alloys display a distribution of equiaxed grain with grain sizes ranging from 20 to 150 μm . The average grain size of the base alloy is about 125 μm . The average grain sizes of the O-2 and N-2 alloys are 101 μm and 95 μm , which is slightly finer than that of the base alloy, respectively. The result indicates that the addition of interstitial atoms can impede the mobility of grain boundaries in the process of annealing. According to the IPF (inverse pole figure) maps, the three alloys show a random orientation without any noticeable preferred orientation. From the inset in Fig. 2, it can be observed that all three alloys have a single-phase bcc structure, which is consistent with the XRD results.

3.2. Hardness and tensile properties

Fig. 3 illustrates the variations in hardness among three alloys. Evidently, the addition of O atoms into TiZrHfNb RHEA results in an increase in hardness from 272 HV to 362 HV, and the addition of N atoms leads to a hardness of 385 HV. It reveals that the strengthening effect of interstitial solid-solution is superior for N than for O.

The tensile true stress-strain curves of RHEAs are shown in Fig. 4. The yield strength of the base alloy is 792 MPa, which is lower than that of O-2 (1125 MPa) and N-2 (1225 MPa) alloys. However, with the addition of O, the yield strength increased significantly and the ductility decreased slightly in the O-2 alloy. Lei et al. [6] reported a similar phenomenon and explained that the addition of oxygen

atoms can form OOCs with matrix elements. The increased ductility of the alloy is attributed to the formation of OOCs, which change the dislocation shearing mode and promote dislocation multiplication. The increased strength is due to the pinning of dislocations by the nanoscale OOCs. However, with the addition of C and N atoms, ordered complexes cannot be found. It can be seen from Fig. 4 that the upper and lower yield points are observed on the curve of the O-2 alloy, so it can be further considered that there are complex interactions between interstitial oxygen atoms and dislocations during tensile deformation.

In addition, the general strengthening mechanisms in Ti-based BCC high-entropy alloys are solid-solution strengthening, grain boundary strengthening, precipitation strengthening, and dislocation strengthening. The calculated yield strength can be expressed in Eq. (2) [31].

$$\sigma_{ys}(\text{Cal}) = \sigma_{ss} + \sigma_{gb} + \sigma_{ds} + \sigma_p \quad (2)$$

In the current study, the phase constitution of three alloys is a single BCC phase, and tension tests are after recrystallized annealed. Therefore, the mechanisms of precipitation strengthening and dislocation strengthening can be ignored. The dominant strengthening mechanisms are solid-solution strengthening and grain boundary strengthening.

One of the key factors contributing to the yield strength of RHEAs is solid-solution strengthening. This type of strengthening is rooted in the interplay between solute atoms and dislocations. In BCC RHEAs, the movement of dislocations is hampered by severe lattice distortions, which are caused by the atomic radius mismatch and the shear modulus mismatch [32]. Toda-Caraballo proposed the solid-solution strengthening model for BCC RHEAs, which can be expressed in Eq. (3) [33].

$$\sigma_{ss} = AGc^{2/3}\delta^{4/3} \quad (3)$$

Table 1

The nominal and actual compositions of three RHEAs (at%).

Alloy	N&A	Ti	Zr	Hf	Nb	O	N
Base	Nominal	25	25	25	25	–	–
	Actual	24.6 $\pm$ 0.8	24.2 $\pm$ 0.6	25.8 $\pm$ 0.1	25.4 $\pm$ 0.2	–	–
O-2	Nominal	24.5	24.5	24.5	24.5	2	–
	Actual	24.9 $\pm$ 1.0	23.8 $\pm$ 0.5	25.3 $\pm$ 0.4	23.6 $\pm$ 0.4	2.4 $\pm$ 0.2	–
N-2	Nominal	24.5	24.5	24.5	24.5	–	2
	Actual	23.8 $\pm$ 0.9	24.2 $\pm$ 0.7	25.5 $\pm$ 0.2	23.9 $\pm$ 1.2	–	2.5 $\pm$ 0.1

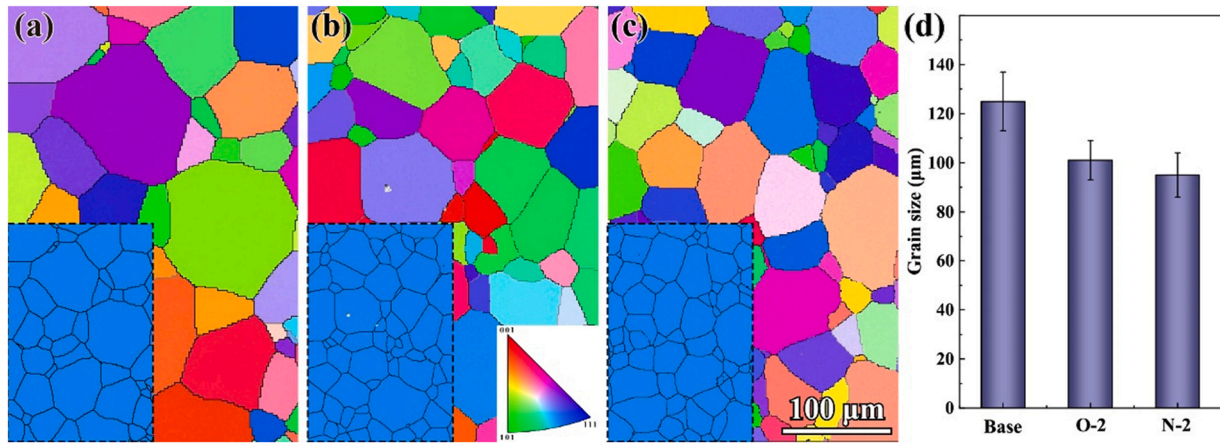


Fig. 2. The EBSD mapping morphologies and phase composition maps of (a) base, (b) O-2 and (c) N-2 RHEAs and (d) their grain size.

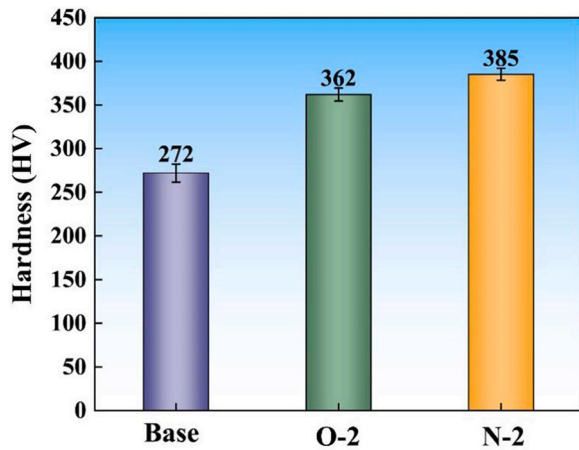


Fig. 3. The hardness of base, O-2, and N-2 RHEAs.

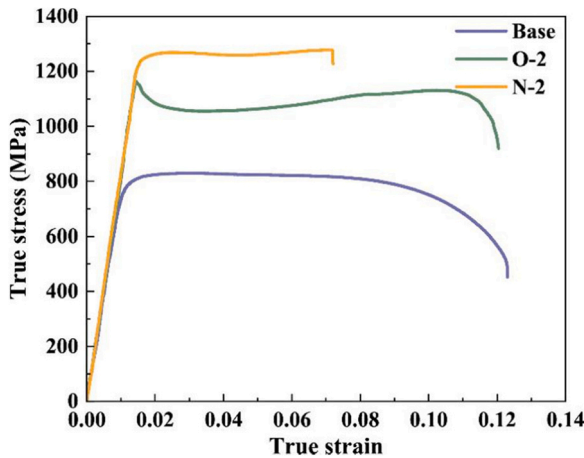


Fig. 4. Tensile true stress-strain curves of the base, O-2, and N-2 RHEAs.

Where A is a scaling constant and has the value of 0.085 in the current work [34], G is the shear modulus of the alloy, c is the atomic fraction of element i , δ term is the mismatch parameter,

Table 2

The parameters of constituent elements, as well as the relative atomic radius mismatch δr_{ij} , and shear modulus mismatch $\delta \mu_{ij}$ (in bold type) of different element pairs.

$i \backslash j, \delta r_{ij}, \delta \mu_{ij}$	Ti	Zr	Hf	Nb
Ti	0	0.286	0.379	0.146
Zr	0.157	0	0.095	0.141
Hf	0.167	0.010	0	0.235
Nb	0.118	0.040	0.050	0
Parameters				
Atomic radius, r (pm)	176	206	208	198
Shear modulus, G (GPa)	44	33	30	38

representing a combination of the shear modulus mismatch and atomic radius mismatch between the solute and solvent, and can be expressed in Eqs. (4) and (5).

$$\delta_i = \xi \sqrt{(\delta \mu'_i)^2 + (\beta \delta r_i)^2} \quad (4)$$

$$\delta \mu'_i = \frac{\delta \mu_i}{1 + |\delta \mu_i|/2} \quad (5)$$

where δr and $\delta \mu'$ represent atomic radius mismatch and shear modulus mismatch; $\xi = 2.5$ in the BCC alloy, which represents a constant depending on the number of active slip systems; $\beta = 3$ in this work, which is about the dominant type of dislocations [35,36].

The strengthening of the solid solution is mainly attributed to lattice distortions resulting from the mismatch of shear modulus and atomic radius. Eqs. (6) and (7) provide the values for the shear modulus and radius mismatch between atoms i and j .

$$\delta r_{ij} = \frac{2(r_i - r_j)}{r_i + r_j} \quad (6)$$

$$\delta \mu_{ij} = \frac{2(G_i - G_j)}{G_i + G_j} \quad (7)$$

where r_i is the atomic radius of pure metal i , and G_i is the shear modulus of pure metal i . The atomic radius and shear modulus of each element, and the calculated value of δr_{ij} and $\delta \mu_{ij}$ are shown in Table 2. The mean values of the shear modulus and atomic radius mismatch in HEAs are provided in Eqs. (8) and (9):

$$\delta r^{ave} = \begin{pmatrix} c_1, c_2, \dots, c_n \end{pmatrix} \begin{pmatrix} 0 & \delta r_{12} & \dots & \delta r_{1n} \\ \delta r_{21} & \ddots & & \delta r_{2n} \\ \vdots & & 0 & \vdots \\ \delta r_{n1} & \delta r_{n2} & \dots & 0 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix} \quad (8)$$

$$\delta \mu^{ave} = \begin{pmatrix} c_1, c_2, \dots, c_n \end{pmatrix} \begin{pmatrix} 0 & \delta \mu_{12} & \dots & \delta \mu_{1n} \\ \delta \mu_{21} & \ddots & & \delta \mu_{2n} \\ \vdots & & 0 & \vdots \\ \delta \mu_{n1} & \delta \mu_{n2} & \dots & 0 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix} \quad (9)$$

Here, assuming that HEAs consist of four principal elements, namely i, j, k , and l , the shear modulus and atomic radius mismatch caused by the i principal element in the HEAs can be determined using Eqs. (10) and (11):

$$\delta r_i = \frac{\delta r_{ijkl}^{ave} - \delta r_{jkl}^{ave}}{\delta c_i} \quad (10)$$

$$\delta \mu_i = \frac{\delta \mu_{ijkl}^{ave} - \delta \mu_{jkl}^{ave}}{\delta c_i} \quad (11)$$

Therefore, by strictly adhering to Eq. (3) and utilizing a combination of Eqs. (10) and (11) with Eq. (4), it is possible to calculate the solid-solution strengthening that is caused by the i element, as demonstrated in Eq. (12):

$$\sigma_{(ss)i} = AGc_i^{2/3}\delta_i^{4/3} \quad (12)$$

As calculated previously, Eq. (13) presents the aggregate solid-solution strengthening generated by every principal element in the alloy.

$$\sigma_{ss} = \sum c_i \sigma_{(ss)i} \quad (13)$$

the calculated values of the solid-solution strengthening are ~680 MPa for the TiZrHfNb alloy.

In addition, previous research [37] reported that under the Fleischer model [38], the strengthening coefficient of the N atom is 377.84 MPa·[at%]^{-1/2}, and the strengthening coefficient of the O atom is 293.23 MPa·[at%]^{-1/2}. With the addition of 2 at% O and N in the base alloy, the calculation values of interstitial strengthening are 417 MPa and 534 MPa, respectively, which also confirms that the

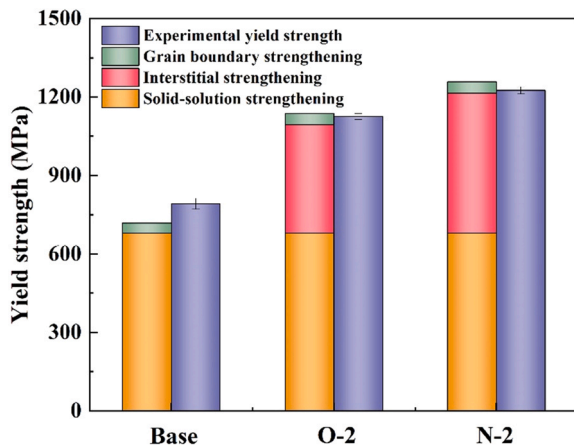


Fig. 5. The comparison of yield strength between experimental and theoretical calculation values in base, O-2, and N-2 alloys.

interstitial solid-solution strengthening effect of N atoms is better than that of O atoms.

The average grain size, as determined from experimental observations, is a key factor in grain refinement strengthening, and can be estimated using the Hall-Petch relationship with the aid of Eq. (14) [39,40]:

$$\sigma_{gb} = KD^{-1/2} \quad (14)$$

where the Hall-Petch constant, denoted as K , holds a value of 430 MPa·μm^{1/2} for the three alloys [40]. Meanwhile, D represents the average grain size.

The overall contribution of σ_{ss} and σ_{gb} on the yield strength of the base, O-2, and N-2 alloys are illustrated in Fig. 5, which demonstrates that solid-solution strengthening primarily contributes to the yield strength of these alloys.

To analyze the fracture morphology of the samples in uniaxial tension, the SEM observations on the fracture surface of the three kinds of RHEAs are shown in Fig. 6. There are a large number of dimples on the fracture surface of these three alloys, which are typical morphologies of ductile fracture. This can indicate that three alloys have good plasticity and good processability at room temperature.

3.3. Wear properties

Fig. 7 shows the evolution of COF during dry sliding wear at temperatures of 25 °C and 500 °C. At room temperature, there is no notable variation observed in the mean COFs of the three specimens, i.e. ranging from 0.61 to 0.65, as shown in Fig. 7a, b, and c. However, the COF curves fluctuate obviously at room temperature because of adhesive wear. At 500 °C, it is evident that a steady-state is reached by each test after an initial running-in period of 400 s. The fluctuation of COF curves is gentler than that at room temperature. The mean values of COFs at 500 °C are decreased to 0.33, 0.34, and 0.28, respectively, which are only 46–55% of that at room temperature. The formation of oxides on the worn surface at high temperatures may lead to the reduction of COF.

The wear rates of three specimens at 25 °C and 500 °C are shown in Fig. 8. With the addition of interstitial atoms O and N, the wear rates tend to decrease at both room temperature and high temperature. At 500 °C, the wear rates of the base, O-2, and N-2 alloys are decreased by 53%, 68%, and 78%, respectively, compared with that at room temperature. Fig. 9 and Fig. 10 show the 3D profiles of wear surface and cross-sectional profiles at 25 °C and 500 °C, respectively. At room temperature, the worn surface 3D profile of the N-2 alloy (Fig. 9c and f) shows a width of ~0.7 mm and a maximum depth of ~8.5 μm, which are apparently lower than those of base and O-2 alloys. At 500 °C, the worn width and depth of three specimens are lower than those at room temperature. Moreover, the worn track of N-2 alloy at 500 °C is the minimum width (0.45 mm) and depth (4.7 μm). The consistent consequences of the wear rate and surface 3D profile provide evidence for the exceptional wear resistance of N-2 RHEA.

The worn surface morphologies of three specimens at 25 °C and 500 °C are shown in Fig. 11. At a temperature of 25 °C, the presence of visible cracks and shallow grooves on the surface of the base alloy suggests the occurrence of abrasion wear and fatigue wear during sliding, as shown in Fig. 11a. Furthermore, the phenomenon of delamination in the wear process is due to the spalling caused by the fatigue of the matrix material, which clarifies the existence of the

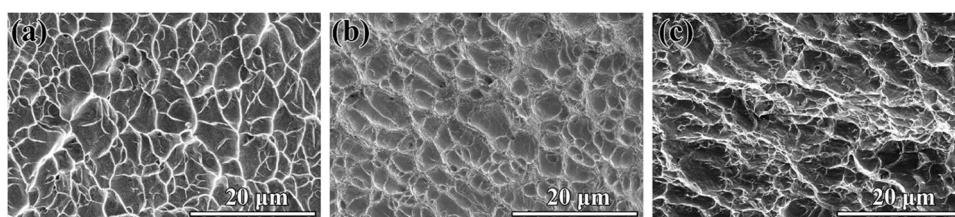


Fig. 6. The fracture surface morphologies of tensile samples: (a) BASE; (b) O-2; (c) N-2.

delamination mechanism. Fig. 11c shows similar cracks and delamination on the surface of the O-2 alloy at 25 °C, indicating that the wear mechanism of the O-2 alloy is fatigue wear and delamination mechanism. On the worn surface of the N-2 alloy, there are shallow grooves and plastic deformation, suggesting that the wear debris forms three-body abrasive wear between the sample and the counterpart ball, as displayed in Fig. 11e. However, the slight grooves reveal a better wear resistance of the N-2 alloy. Furthermore, EDS analysis was carried out on the dark and light regions of the wear morphology of the three samples at room temperature, as shown in Table 3. It revealed a noteworthy difference in the oxygen levels between the dark and light regions, implying diverse degrees of oxidative wear on the worn surfaces of the three alloys at ambient conditions. EDS analysis detected the presence of silicon on the worn sample surface, indicating material transfer between the samples and the counterpart balls, indicative of adhesive wear at ambient temperature. At 500 °C, the wear surface of the sample is smoother than that at room temperature. There are no obvious grooves, cracks, and spalling on the worn track surface of the three samples, as shown in Fig. 11b, d, and f. Table 4 displays EDS-detected oxygen

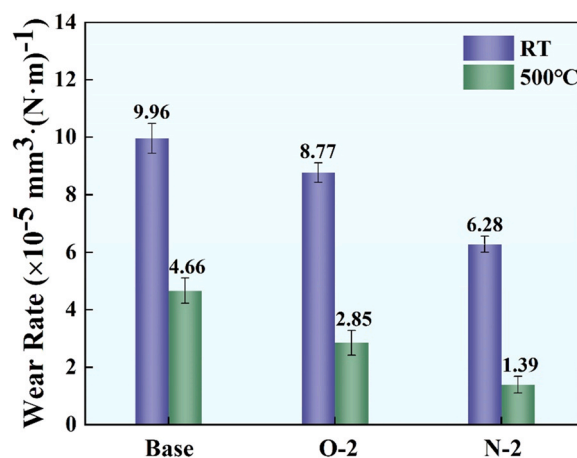


Fig. 8. The wear rates of three kinds of RHEAs at 25 °C and 500 °C.

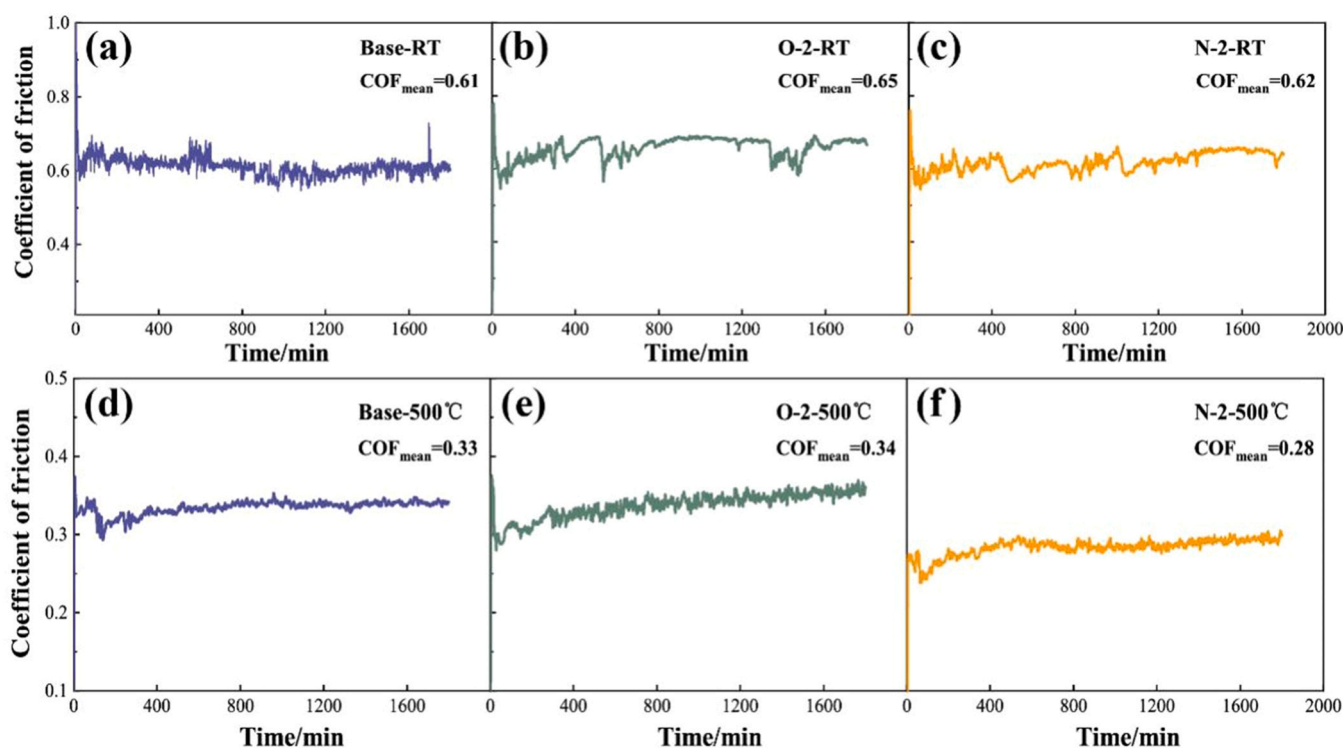


Fig. 7. The COF curves of three kinds of RHEAs at room temperature and 500 °C: (a)&(d) Base; (b)&(e) O-2; (c)&(f) N-2.

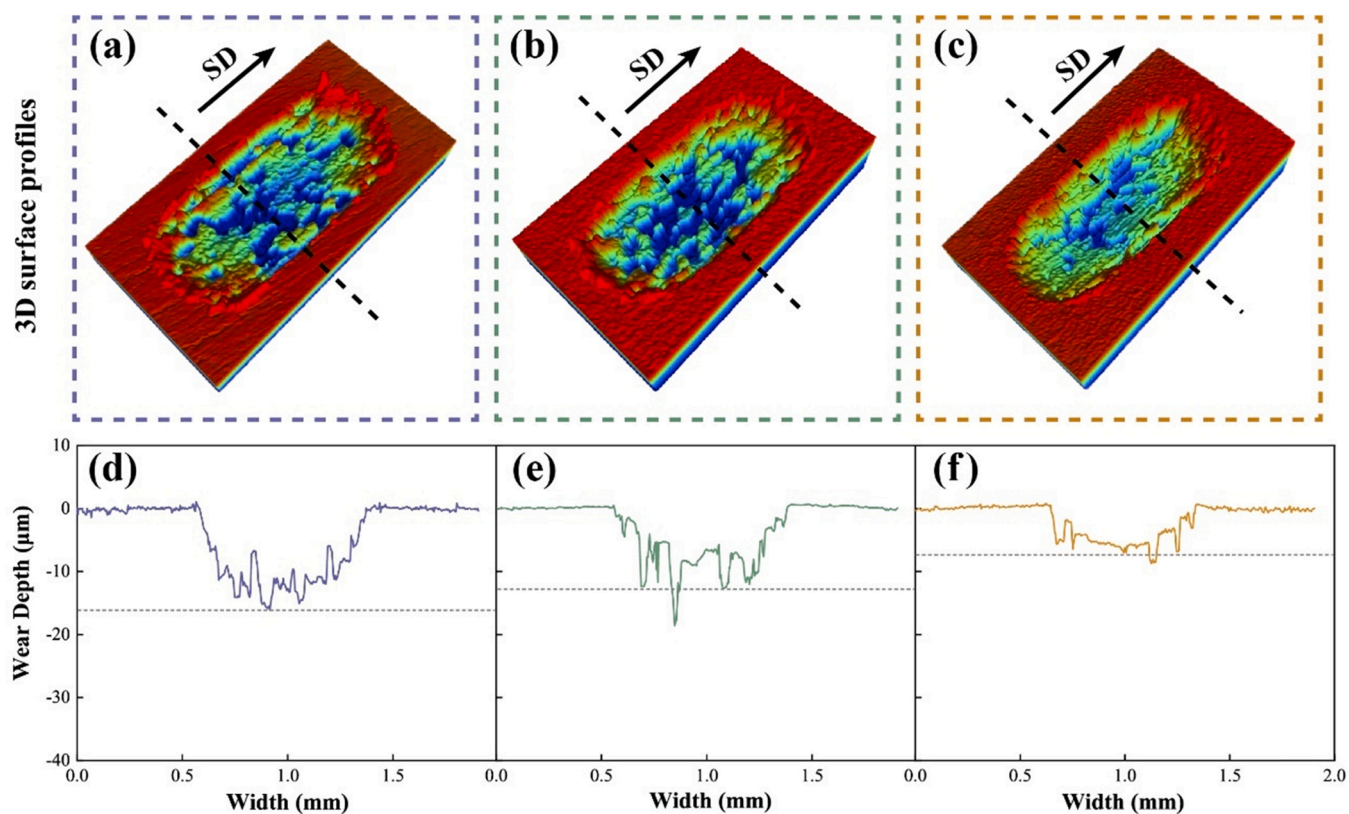


Fig. 9. The 3D profile of the wear surfaces and cross-sectional profiles at room temperature: (a)&(d) Base; (b)&(e) O-2; (c)&(f) N-2.

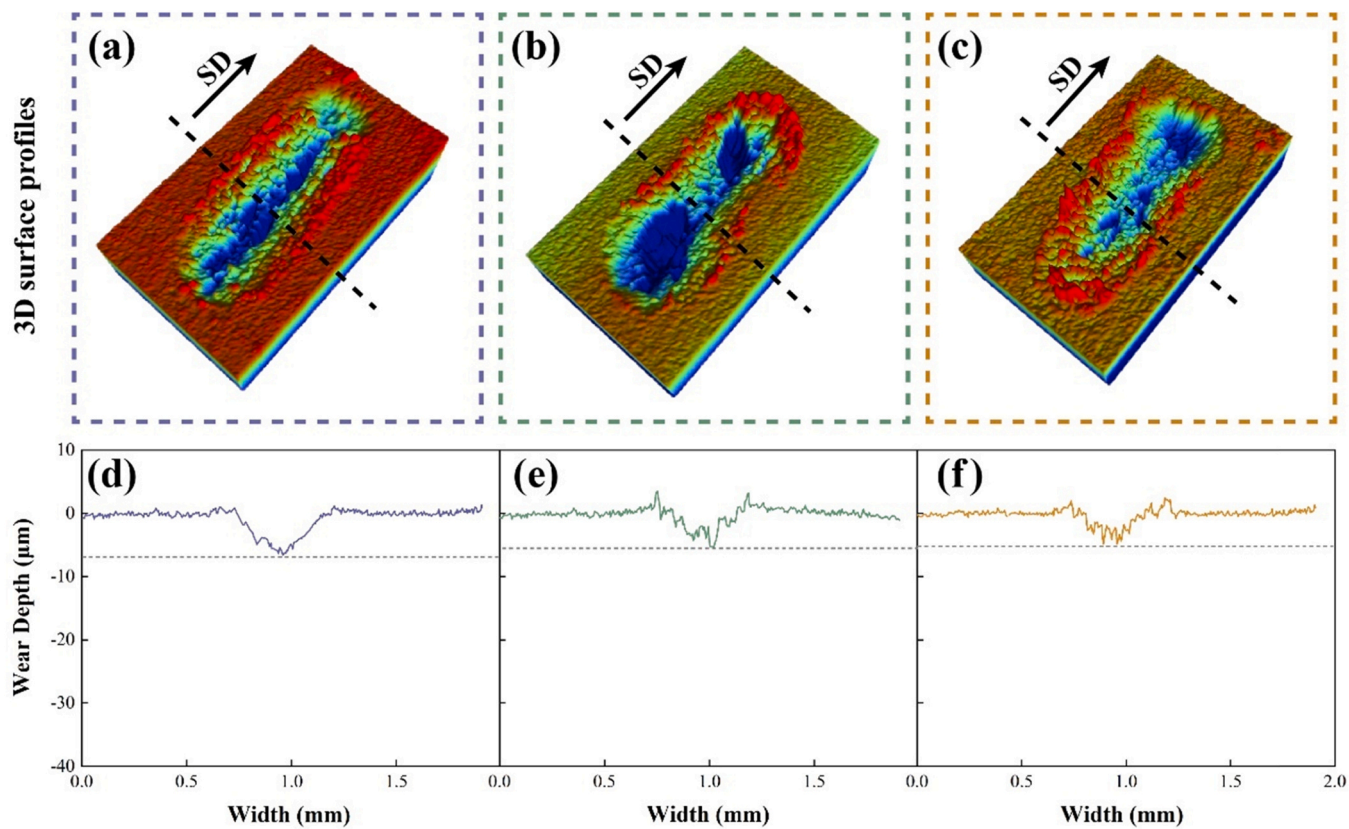


Fig. 10. The 3D profile of the wear surfaces and cross-sectional profiles at 500 °C: (a)&(d) Base; (b)&(e) O-2; (c)&(f) N-2.

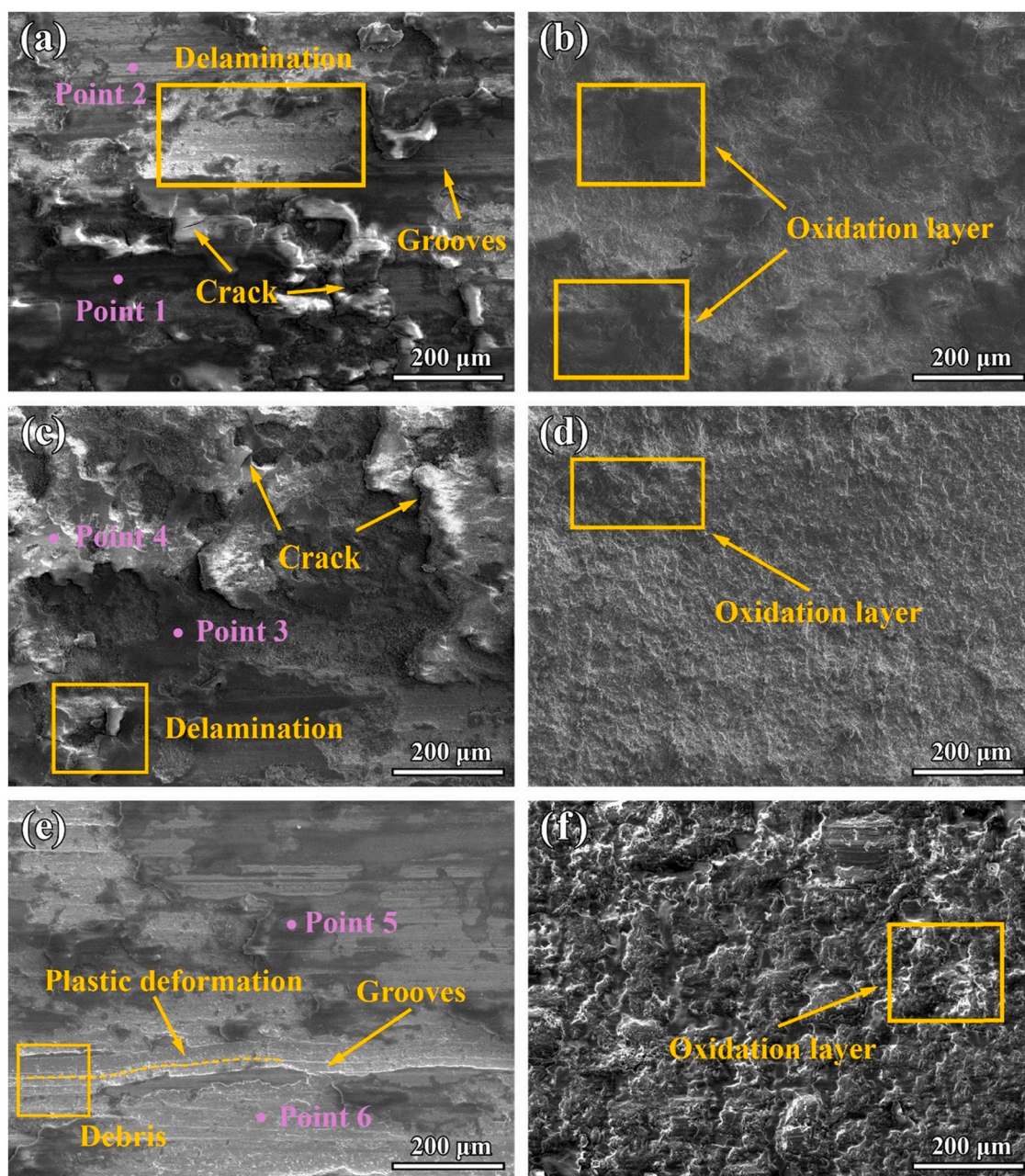


Fig. 11. The surface morphologies of wear tracks of three specimens at 25 °C and 500 °C: (a) BASE alloy at 25 °C; (b) BASE alloy at 500 °C; (c) O-2 alloy at 25 °C; (d) O-2 alloy at 500 °C; (e) N-2 alloy at 25 °C; (f) N-2 alloy at 500 °C.

Table 3

Chemical compositions of worn surfaces at 25 °C. (at%).

Elements	Base		O-2		N-2	
	Point 1	Point 2	Point 3	Point 4	Point 5	Point 6
O	69.92	25.84	70.71	35.93	70.18	16.47
Si	10.75	-	14.30	2.15	11.96	1.19

content on the wear track surfaces of the three samples, revealing a significantly higher concentration at high temperature compared to ambient temperature. This result reveals that the wear mechanism is dominated by oxidative wear at 500 °C. However, there is no obvious

Table 4

The chemical composition of worn surface at 500 °C. (at%).

Elements	Base	O-2	N-2
O	67.83	71.26	69.48
Si	-	0.23	-

silicon on the worn surface by EDS analysis, indicating a seldom occurrence of adhesive wear at high temperature. A surface oxide layer that possesses friction-reducing properties has been observed to form on the sample. At 500 °C, the presence of an oxide layer on

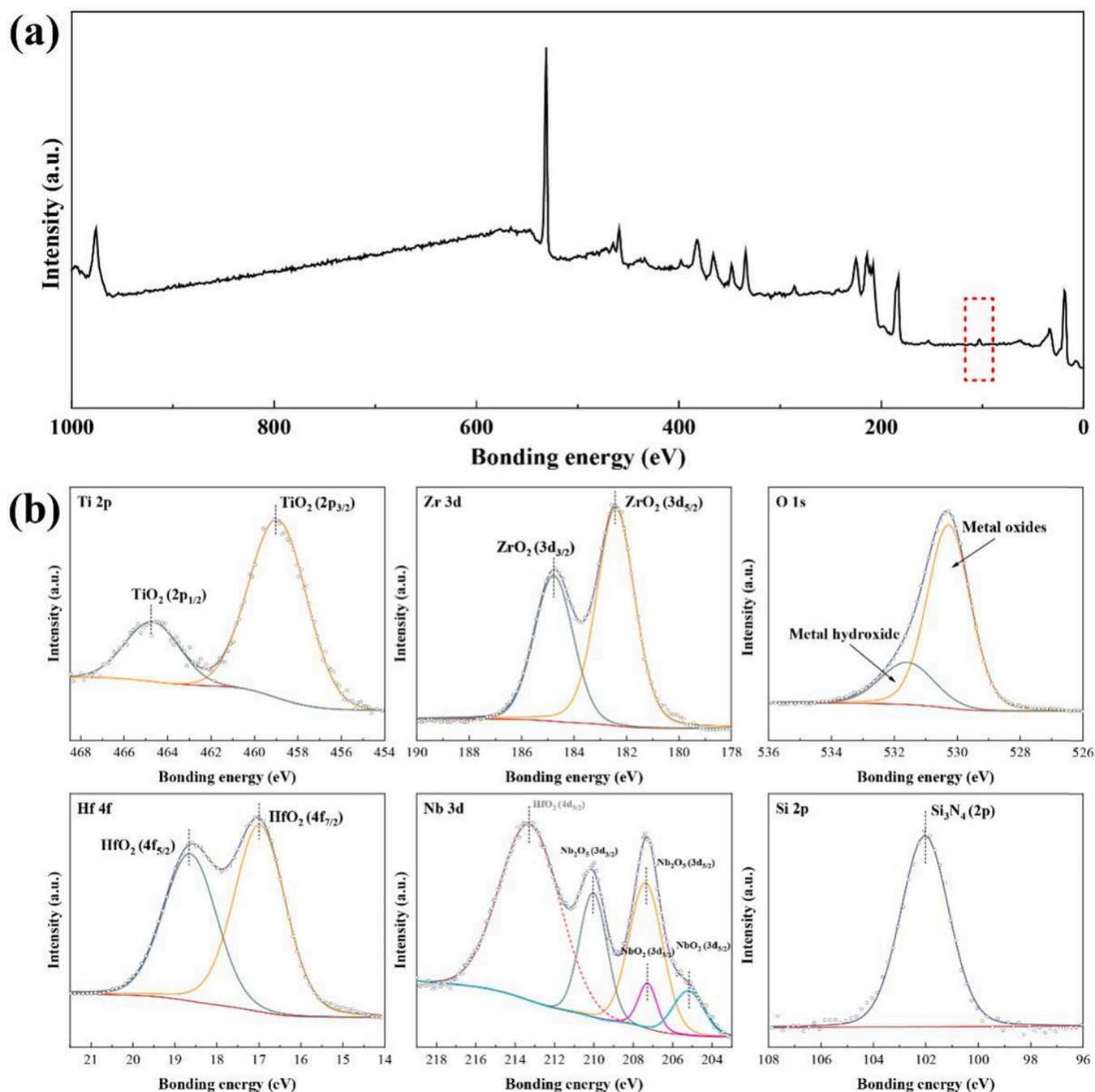


Fig. 12. The XPS analysis on the worn surface of N-2 alloy at 25 °C: (a) survey spectra and (b) high-resolution spectra.

the surface of the alloy results in a marked reduction of the adhesive wear, leading to a significantly lower wear rate compared to that observed at 25 °C.

To further investigate the adhesive wear and oxidative wear on the worn surface of N-2 alloy, the XPS analysis was carried out to distinguish the transformation between samples and counterparts, and oxidation products during dry slide wear. The survey spectra and the high-resolution XPS spectra on the worn surface of the N-2 alloy at 25 °C and 500 °C are shown in Fig. 12 and Fig. 13. There are obvious

peaks of metal oxidation in the XPS survey spectra of the N-2 alloy worn at 25 °C and 500 °C. It is worth noting that the Si 2p peak clearly occurred in the XPS survey spectrum of N-2 alloy worn at 25 °C, as shown in Fig. 12a. However, at 500 °C, the Si peaks are not detected in the survey spectrum. It is suggested that there is severe adhesive wear at 25 °C, compared with that at 500 °C. These results are completely consistent with the EDS results.

The various oxides of the worn surface at 25 °C and 500 °C are analyzed by XPS high-resolution spectra, as shown in Fig. 12 and

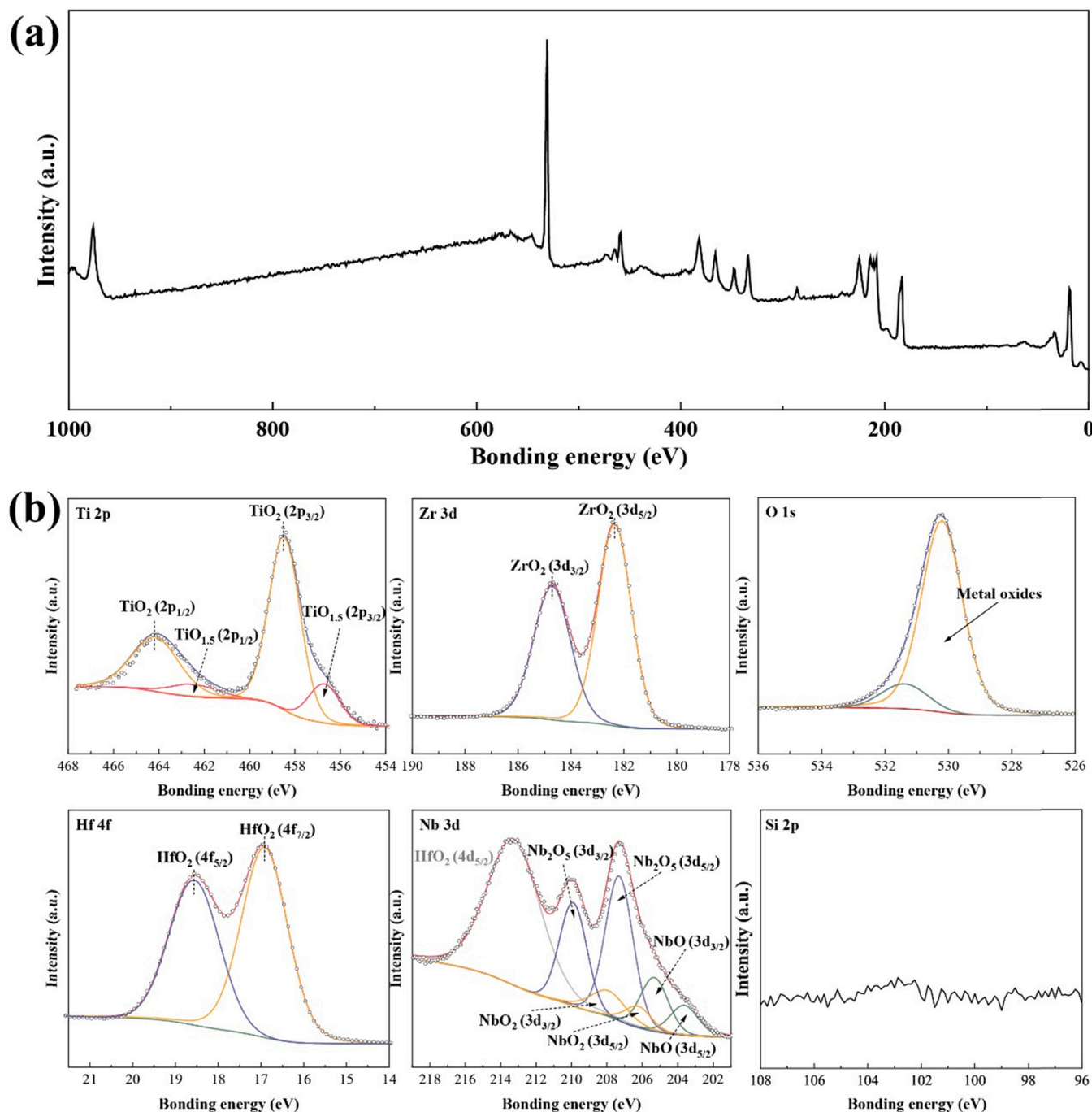


Fig. 13. The XPS analysis on the worn surface of N-2 alloy at 500 °C: (a) survey spectra and (b) high-resolution spectra.

Fig. 13. The oxides are significantly different between 25 °C and 500 °C. At room temperature, the oxides consist of TiO_2 , ZrO_2 , HfO_2 , NbO_2 , and Nb_2O_5 . With increasing temperature, the oxides are complex, and the new oxides, Ti_2O_3 and NbO , are formed on the worn surface at 500 °C, as displayed in Fig. 13, indicating that oxidation wear is more severe than that at 25 °C.

Cross-sectional characterizations of the wear track are critical to analyze the dry sliding wear behavior. TEM techniques were employed to detect the normal direction-sliding direction (ND-SD) cross-sectional microstructure of the wear track formed at 500 °C for the N-2 alloy, as depicted in Fig. 14 and Fig. 15. It is interesting that the microstructure of the cross-section is delaminated about three

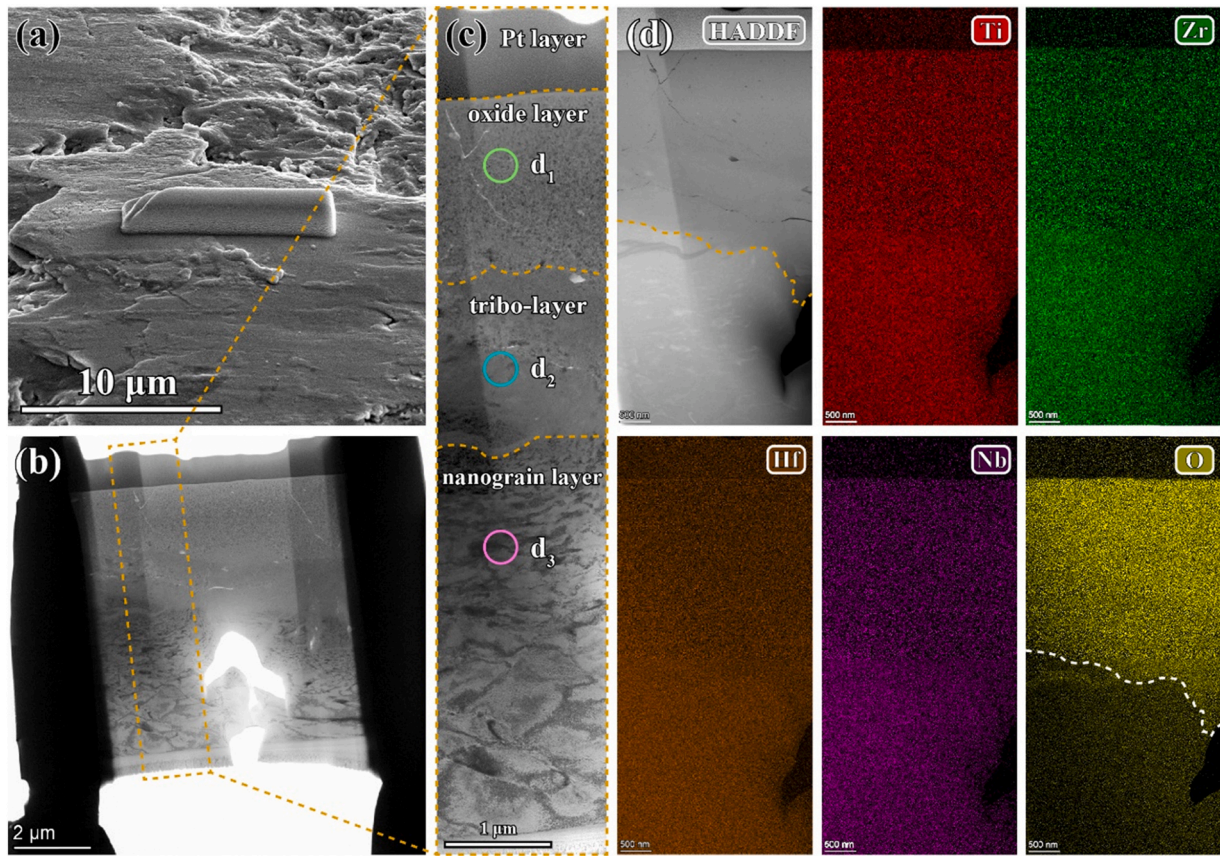


Fig. 14. The cross-sectional characterizations of the N-2 RHEA after dry sliding at 500 °C: (a) SEM image of the target region protected by a platinum layer; (b) TEM image of the lamella milled by FIB; (c) a bright-field TEM image showcasing the cross-sectional microstructure beneath the sliding surface; (d) a HAADF-STEM image accompanied by corresponding EDS elemental maps.

layers from the worn surface to the subsurface (Fig. 14c): (1). The top oxide layer formed in situ during the wear process; (2). The tribo-layer is the result of wear debris undergoing spalling and compaction; (3). The nanograin layer is formed by friction stress/strain under the cyclic sliding load.

The composition of each layer can be determined by selected area electron diffraction (SAED) and EDS. The top layer is a nanoscale oxide with the grain size about 13 nm, including oxides of Ti, Zr, Hf, and Nb, as shown in Fig. 14d and Fig. 15a, d, f. The fraction of nano-oxides in the middle tribo-layer is significantly reduced, and there are deformed and elongated grains at the same time, indicating that the friction stress results to the deformation of grains and the surface oxides are peeled off and then compacted, as shown in Fig. 15b and e. At the bottom layer, there are ring patterns of the BCC matrix, but they are composed of ultra-fine grains, indicating that friction induces the grain refinement of the subsurface, as shown in Fig. 15c. The excellent wear resistance of

N-2 alloy at high temperature can be attributed to the following reasons: on the one hand, the nano-oxides generated in situ on the surface layer play a lubricating role. On the other hand, the boundary strengthening of ultra-fine grains can further resist wear damage of the subsurface. The multi-layer Synergistic effect improves the wear resistance of N-2 alloy.

In summary, at room temperature, the wear rate decreases with the addition of O and N in the TiZrHfNb alloy. The main wear mechanisms at room temperature are severe abrasive wear, adhesive wear, and partial oxidative wear. At 500 °C, the wear rate also decreased with the incorporation of O and N into the base TiZrHfNb alloy. There is a notable reduction in the wear rate at high temperatures compared to ambient temperatures, which can be attributed to a change in the wear mechanism. The formation of surface oxides plays a critical role in reducing the adhesive wear. Therefore, the predominant wear mechanisms are oxidative wear and slight abrasive wear at 500 °C.

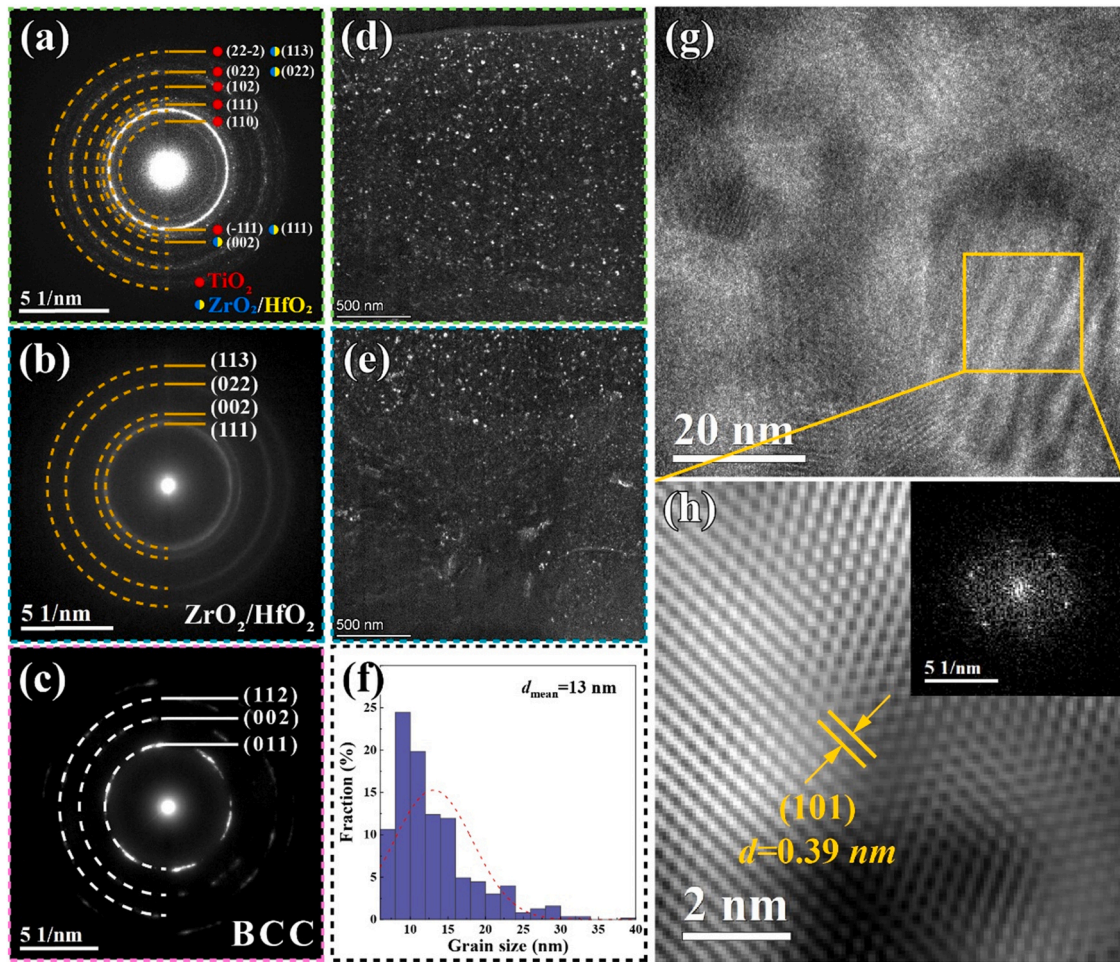


Fig. 15. The cross-sectional TEM characterizations of the N-2 alloy after dry sliding at 500 °C: (a) (b) (c) are SAED ring patterns of oxide layer, tribo-layer, and nanograin layer, respectively; (d) & (e) are dark-field TEM images of the oxides layer; (f) is the grain size distribution of the oxides in the oxide layer; (h) shows the inverse fast Fourier transform (IFFT) image of the selected regions, as indicated in (g); the inset displays the fast Fourier transform (FFT) pattern of the chosen region.

4. Conclusion

In this study, we investigated the microstructures, mechanical properties, and wear behavior of the three alloys, base, O-2, and N-2. The primary conclusions are as follows:

- (1) The phase constitution of three alloys are the single BCC structure. With the addition of interstitial atoms O and N, the grain size of the base alloy is refined from 125 μm to 101 μm (O-2) and 95 μm (N-2).
- (2) Doping interstitial atoms leads to a substantial enhancement in the yield strength and hardness of O-2 and N-2 alloys. The N-2 alloy exhibits the highest yield strength (1225 MPa), which is predominantly attributed to solid-solution strengthening and grain strengthening mechanisms. On the other hand, the O-2 alloy displays a remarkable balance between strength and ductility.
- (3) At ambient conditions, the wear rate of three alloys is higher compared with that at 500 °C. The primary wear mechanisms at room temperature include severe abrasion, adhesive wear, and oxidative wear, whereas at 500 °C, oxidative wear and slightly abrasive wear dominate. The formation of an oxide film on the worn surface can reduce the COF. The N-2 alloy demonstrates improved wear resistance due to the synergistic effect of oxide layer, tribo-layer, and nanograins layer.

CRediT authorship contribution statement

Chi Jin: Investigation, Data curation, Formal analysis, Validation, Writing – original draft. **Xiaolin Li:** Investigation, Writing – review, Supervision, Funding acquisition. **Junhong Kang:** Data curation, Validation. **Haozhe Li:** Data curation, Validation. **Haifeng Wang:** Writing – review & editing, Supervision, Funding acquisition.

Data availability

Data will be made available on request.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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